PHY202 Practice Exam on chapter(s): 16, 17

Name:____

These questions assess your understanding of the material from Chapters 16, 17 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. Calculate the sound intensity level in decibels for a sound wave traveling in air at 0.70°C with a pressure amplitude of $0.0366\ \mathrm{Pa}$. Recall that the density of air is $1.29\,\frac{\mathrm{kg}}{\mathrm{m}^3}$ and the speed of sound in air at 0°C is $331\,\frac{\mathrm{m}}{\mathrm{s}}$.
   (Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ex17_2)

   ANSWER: ____________________________

2. Suppose that a 2.64×10^3-kg car has a suspension system with a force constant $1.40 \times 10^4\,\frac{\mathrm{N}}{\mathrm{m}}$. If the car hits a bump and bounces with an amplitude of $3.2\ \mathrm{cm}$, then what is its maximum vertical velocity assuming no damping occurs?
   (Chapter 16, Section (5) Energy and the Simple Harmonic Oscillator - cp1e_ex16_6)

   ANSWER: ____________________________

3. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the period of oscillations for a car with shoddy shock absorbers if the car's total mass is $1.09 \times 10^3\ \mathrm{kg}$ and the force constant of the suspension system is $1.93 \times 10^4\,\frac{\mathrm{N}}{\mathrm{m}}$.
   (Chapter 16, Section (3) Simple Harmonic Motion - cp1e_ex16_4)

   ANSWER: ____________________________

4. Suppose you have a tube that is closed at one end. Its length is $9.1\ \mathrm{cm}$ and the air temperature is 0.21°C. What is the frequency of its second overtone?
   (Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ex17_5b)

   ANSWER: ____________________________

5. When Sarah gets in her car, it lowers by $1.0\ \mathrm{cm}$. If Sarah's mass is $98.\ \mathrm{kg}$, then what is the force constant of her car's suspension system?
   (Chapter 16, Section (1) Hooke's Law - cp1e_ex16_1)

   ANSWER: ____________________________

6. How much energy is stored in the spring of a toy gun that has a force constant of $51.4\,\frac{\mathrm{N}}{\mathrm{m}}$ and is compressed $24.8\ \mathrm{cm}$?
   (Chapter 16, Section (1) Hooke's Law - cp1e_ex16_2)

   ANSWER: ____________________________

7. A medical imaging device produces ultrasound by oscillating with a period of $0.27\ \mu\mathrm{s}$. What is the frequency of this oscillation?
   (Chapter 16, Section (2) Period and Frequency in Oscillations - cp1e_ex16_3)

ANSWER: ______________________________

8. A sonar echo returns to a submarine $6.99\ \mathrm{s}$ after being emitted from the submarine. What is the distance from the submarine to the object, given that the speed of sound in this water is $1.50\ \frac{\mathrm{km}}{\mathrm{s}}$?
(Chapter 17, Section (2) Speed of Sound, Frequency, Wavelength - cp1e_17.8)

ANSWER: ______________________________

9. Suppose you have a tube that is closed at one end. If you want its fundamental frequency to be $290.\ \mathrm{Hz}$ and the air temperature is 0.87□°C, then what should the length of the tube be?
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ex17_5a)

ANSWER: ______________________________

10. What is the length of a pendulum that has a period of $0.53\ \mathrm{s}$?
(Chapter 16, Section (4) Simple Pendulum - cp1e_problem16_22)

ANSWER: ______________________________

11. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of $958.\ \frac{\mathrm{kg}}{\mathrm{m}^3}$ to muscle tissue with a density of $1.027 \times 10^3\ \frac{\mathrm{kg}}{\mathrm{m}^3}$. The speed of ultrasound in the fat is $1.42 \times 10^3\ \frac{\mathrm{m}}{\mathrm{s}}$ and the speed of ultrasound in the muscle tissue is $1.58 \times 10^3\ \frac{\mathrm{m}}{\mathrm{s}}$.
(Chapter 17, Section (7) Ultrasound - cp1e_ex17_7)

ANSWER: ______________________________

12. Suppose a 0.302-kg object is connected to a horizontal spring. There is friction between the object and the surface, and the coefficient of kinetic friction μ_k is 0.19. What total distance does the object travel if it is released from rest at a distance of $14.\ \mathrm{cm}$ from its equilibrium position? The force constant of the spring is $72.3\ \frac{\mathrm{N}}{\mathrm{m}}$.
(Chapter 16, Section (7) Damped Harmonic Motion - cp1e_ex16_7)

ANSWER: ______________________________

13. Suppose a train sounding its $209.\text{-Hz}$ horn is moving at $18.5\ \frac{\mathrm{m}}{\mathrm{s}}$. What frequency does a person standing by the tracks hear as the train approaches? Given this particular day's temperature, the speed of sound is $340.\ \frac{\mathrm{m}}{\mathrm{s}}$.
(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ex17_4)

ANSWER: ______________________________

cp1e: College Physics (1st Edition) by OpenStax
PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway
tkd: I did not refer to anything while writing this question